KAAVYA DHIR

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GitHub — LinkedIn — LeetCode

Summary

Computer Science undergraduate experienced in building and deploying AI/ML and backend systems using Python, C++, SQL, FastAPI, React, and PyTorch. Hands-on with REST APIs, RAG, data analysis, Docker, AWS, and CI/CD. LeetCode contest rating 1743 (top 10.17% globally).

Education

Thapar Institute of Engineering and Technology 2023 – 2027
B.Tech. Computer Science and Engineering CGPA: **7.40**

Delhi Public School, Meerut 2022 – 2023
Class XII Percentage: **90%**

Delhi Public School, Meerut 2020 – 2021
Class X Percentage: **98%**

Technical Skills

Languages	Python, C++, C, JavaScript, SQL
AI & ML	Google Gemini API, RAG, text embeddings, prompt engineering, PyTorch, NumPy
Backend & APIs	FastAPI, REST APIs, Pydantic, Node.js, Express.js
Frontend	React.js, HTML, CSS, Tailwind CSS
Cloud & DevOps	AWS EC2, Linux, Docker, HTTPS/TLS, GitHub Actions, Git, pytest
Databases & Data	SQL, SQLite, MySQL, MongoDB, Pandas
Core CS	DSA, OOP, DBMS, Operating Systems, Computer Networks, System Design

Projects

Prior Authorization Copilot [GitHub] [Live] *Python — FastAPI — Gemini — Docker — AWS EC2*

- Built a healthcare decision-support API that evaluates doctor's notes against insurance policies and returns **approve, deny, or escalate** with cited evidence.
- Separated LLM-based policy evaluation from a deterministic **Python rule engine**, requiring every criterion to be met for approval.
- Deployed on **AWS EC2** using Docker and HTTPS with **37 tests**; deleting one documented fact moves the same request from **approve at 73%** to **escalate at 49%**.

Denial Intelligence [GitHub] [Live Report] *Python — SQL — NumPy — CI/CD*

- Analyzed US health-insurance data covering **5,424 plans and 326 insurers** to identify abnormal claim-denial patterns.
- Used SQL and **median-based statistics** instead of averages to detect outliers within each market — caught **6 of 6** planted cases where the standard method caught **0 of 6**.
- Combined denial and appeal-overturn signals to identify corroborated findings; validated with **86 tests** and CI.

HumanMotion AI [GitHub] *React.js — FastAPI — PyTorch — T2M*

- Built a full-stack app that converts natural-language prompts into **3D human motion** using a pretrained T2M model.
- Integrated the research model with **FastAPI and React**, building a 3D skeleton viewer and downloadable motion outputs.
- Found **37s of a 48s request** was model-loading overhead, identifying an in-memory fix for roughly **4x lower latency**.

Experience

R&D Member, Echoes Society 2023 – 2025
Thapar Institute of Engineering and Technology

- Researched and evaluated proposals for society initiatives, and produced the **documentation, reports and research material** the R&D team worked from.
- Planned and coordinated initiatives with the team, tracking tasks and deliverables through to execution.

Certifications & Achievements

- **LeetCode Contest Rating 1743** — top **10.17%** globally (overall rank under 600,000).
- **Runner-Up**, BIOS Hackathon, Thapar Institute of Engineering and Technology (November 2025).
- **Dynamic Programming Camp**, AlgoUniversity — Matrix Exponentiation, SQRT Decomposition [Certificate].